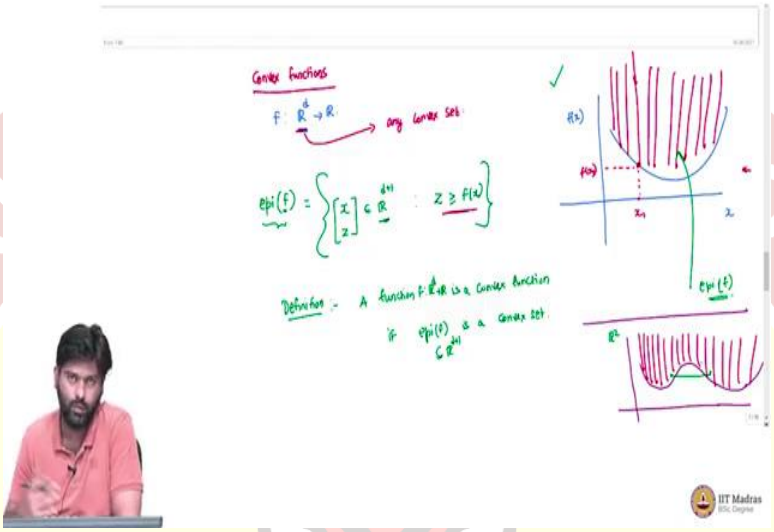


IIT Madras

ONLINE DEGREE

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Convex Functions

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The next thing that we want to look at is the definition of what do we mean by a convex function? So, we have seen what are convex sets, now we need to understand what are convex functions and because we want to eventually in machine learning think of the goodness of whatever that we are outputting and the goodness will often be measured using some kind of a performance function and more often than not in machine learning we will encounter functions which are of convex nature.

For a particular reason that we will see in this course in by the end of this lecture but before that we need to understand what do we even mean by saying that a function is a convex function. Now there are several alternate definitions for thinking about function as a convex function, we will see couple of these definitions and it is important to understand all these definitions because it gives you more you know complete perspective of what convex functions are, and how to recognize a function as convex.

If tomorrow, you have to prove a function is a convex function it would be good to have in your you know armor all possible ways of attacking the problem. So, that is why we will give multiple definitions of convex functions, okay. So, the one of the easiest way or one of the ways that will

easily build from whatever we have already seen which is the notion of convex sets is by understanding the following definition.

So, we will think of a function of in general as Rd to R any function that takes d inputs and then spits out a real number but it is always easy to visualize this in two dimension or one dimension in this case R to R functions. So, think of this as x and this is f(x) and we somehow want to argue that maybe this function is convex or not function or not convex. So, we need to kind of give a definition for what do we mean by a convex function.

We know what are convex sets, we need to define convex functions. So, we are going to think of it as Rd but in general you can also think of this can be any convex set. So, maybe you are looking at a function which is restricted to a particular domain and that domain has to be a convex set; so your functions value on that domain now you can ask whether your function is a convex function or not?

If the domain itself is not convex then the function is not by definition a convex function and if it is Rd then naturally this is a convex set and so that problem of thinking about whether the domain is convex or not does not arise in this case. But okay, so but there are several functions from Rd to R not all of them are convex we still have to define what do we mean by saying a function is convex, here is one way to define this.

Now we are going to define; given a function, you give me a function f I am going to define a set out of this function, so because I only understand convexity of sets at this point so I want to understand I want to give a definition of a convex function via the notion of convex sets that we have already understood and so given a function I am going to define a set out of this and that set is what I will call as an epi graph, in short epi is just a set.

So, think of this as a subset of Rd in fact this we will see it is not a subset of Rd it will be Rd + 1 but let me define this okay; so what is an epi graph? An epi graph is the set of all points of the form x and z so it is a point in d + 1 dimension so this is in Rd + 1; now what is so special about this x and z?

Well, you should have somehow you have to relate it to f so because this is a set that is associated with f and the way we will associate with it is by saying that we will consider all those points x ,

z where z is greater than or equal to f(x). What do I mean by this? So let us, this means that if I am given a function like this now I take any x and I want to look at which are the points that I am considering in the epi graph of f.

Well, it is going to be the set of points, now I should think of this as a two dimensional plane where I will go x in this direction and in the vertical direction I have to go at least f(x). So, anything above or equal to f(x) is okay. So, which means that this would be f(x), so this point would be in my epigraph not just this point this whole line so this is not going to be there.

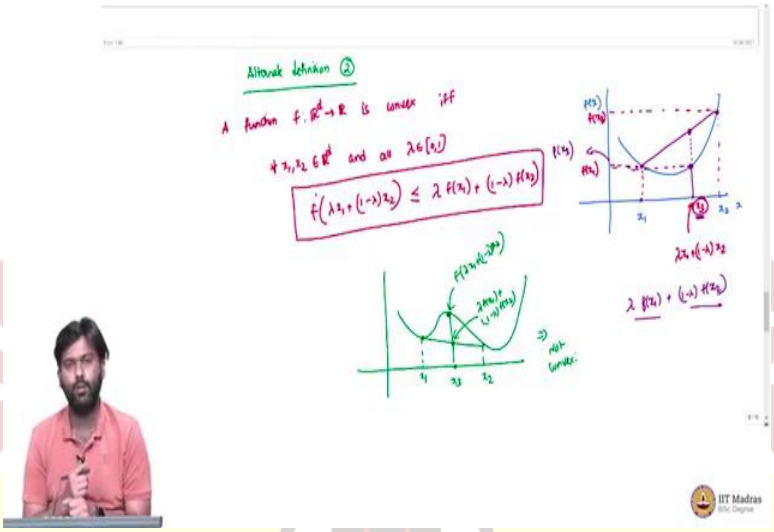
So, this was just to show that this is at f(x), so this is x 1 let us say f(x) 1. So, now all these points this is like an infinitely extending line but yeah so this is nevertheless this line. So, now I do this for every point which means that the set that I am associating with this function is this set so now this is the epi graph of f; it is a set in Rd + 1 because your x itself is in Rd and the value that x takes is f(x) which is a number so you add the text or dimension.

So, the definition now then is we will say a function f, so we will say a function f is convex if is called when I say is convex is a convex function that is what I mean if the epi graph of f is a convex is called a convex function so this is a definition. So, given a function I compute the epi graph of the function and if I see that the epigraph is a convex set then I will call the function a convex function.

So, this function again let me make this clear this is Rd to R it is an epigraph is in Rd + 1. So, this is a convex function, so this is this you can of course I need to specify what the f is but pictorially at least it looks like this is a convex function. Again, pictorially speaking what might not be a convex function, well a function which may be looks like this, so maybe it gets extended after that.

So now is this a convex function? Well, not really because now the epigraph is going to be all these points, now so the set is the epigraph set is not a convex set because now I can take two points in the epigraph set associated with ref and because the line joining those two points as now you have to think of this as a in R2, so this is R2 this line joining this in R2 the line segment does not entirely sit inside the epigraph of f in this particular case. And so this is not a convex set and so this is not a convex function also. So, this is one definition of convex functions but there are other definitions as well and that is what we will see next.

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Okay, so here is a very interesting and relevant alternate definition of convex functions, alternate definition let us call this definition 2 so we already had definition 1. And again, this can be easily pictorially visualized. You have a function like this and you want to ask if this is a convex function well let me take two points x 1 and x 2 and see where their function evaluations are.

So, this is f(x 1)this is f(x) 2. Now if I take a point between x 1 and x 2, so we know how to get points between x 1 and x 2 it is just λ times x 1 + 1 - λ times x 2 for some λ; let us say I pick it for some λ here so this is λ x 1, x 1 - λ x 2 let us call this x 3 perhaps.

Now I see how is the functions value at x 3, so now if I look at this point which is λ which is f(x 1)and this point which is f(x 2)now if I use the same combination but not for the data point but for the functions value so which means that if I look at λ times f(x 1) + 1 - λ times f(x) 2.

So, if I look at that value and now as I, if I think of that as a function of λ now how is that going to look like; well that will be simply you can think of it as when λ is 0 it is going to be well λ is 1 it is going to be f(x 1) and if λ is 0 it is going to be f(x 2)and in between it is going to kind of linearly interpolate between these two values. So, and that would be something like this.

So, now for a new x 3 if I look at the corresponding value on this line segment that interpolates between you know f(x 1)and f (x 2) now that value would actually be higher than the value of f(x

3) itself. So, the functions evaluation at x 3 which is this value, so this is maybe f(x 3) is less than in this case λ times f(x 1)+ 1 - λ times f(x 2) .

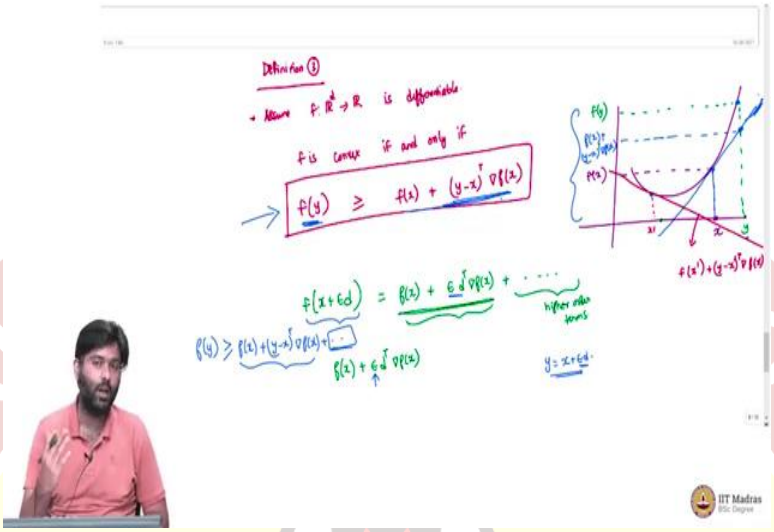
So, now if f is a convex function then it necessarily has to be this case which gives us our alternate definition, a function again for simplicity let us think of this as Rd to R is convex if and only if for all x 1, x 2 in Rd and all λ in 0 , 1 if it happens to be the case that f of λ x 1 + 1 - λ x 2 which is the functions evaluation at this point x 3 in this case we called it x 3 is less than or equal to λ times f(x 1)+ 1 - λ times f(x2) .

So, this is another way to define convex functions and one can show that this is exactly equivalent to our previous definition as well. Now just to confirm that you might have a function of this form which is like this, so now which means that if I take two points and then linearly interpolate maybe I look at this point which is x 1 maybe this point this is x 1, this is x 2 and maybe this is x 3. So, the function's value at x 3 is actually higher than the function's value of the interpolation of x 1 and x 2 λ of f(x 1)and f(x 2) .

So, this is λ f(x 1)+ 1 - λ f(x 2)whereas this is f of λ x 1 + 1 - λ x 2 and in this case it happens that this is higher than the linear quantity and so this cannot be a convex function, so which implies not convex.

So, in some cases finding the epigraph and trying to prove the epigraph is a convex set might be a little harder whereas you can probably apply this definition and exhibit certain points directly to show that a function is not convex or you can show that for general x 1 and x 2 if you can argue that this holds then well that is another way to say that the function is also convex. What we will do next is look at another alternate definition of convexity which is again very useful.

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Okay, so this is the third definition of convexity, so this is definition 3 and for this definition we are going to assume that our function is differentiable, so is differentiable which means that the gradient of function a exists at every point. So, now assume f is differentiable now we want to argue; what can we argue? We want to say that f is convex if and only if the following holds.

So, you take any point y and you compare it with f(x) + y - x transpose del f(x) now it will be the case that f y would be greater than or equal to this. So, this needs some explanation but this is the definition. What are we saying here, we are saying here that we take a function let us again look at a simple function? So, let us say this is the function and maybe this is my x now I want to compute of course this is f(x) that is fine.

I want to compute f of y at some other point now let us say at this point. So, this is maybe y, now or y could be on the other side also so does not I think it is easier to look at the other side signs okay. So, this is this is actually f of y, this is f of y. Now what does this inequality say; it says that this value f of y is actually greater than something.

So, the left hand side is just this value which is this quantity f of y but what is it greater than? Well, it is greater than this quantity which is f(x) + something which is what which is you look the gradient of the function at the point x and then look at the y - x transpose del f(x) which means that you are essentially thinking of this function as a linear function around this point x.

So, which means that if you actually plot this if you think of x as fixed and as a function of y if you plot this think of this as a function then that would look like something like this, it is going to look like this and now if you ask what is the value at a particular y that we have that value is going to be here. So, now what are we saying; we are saying that this quantity here so this is f(x) + y - x transpose del f(x).

Of course, there is no transpose or anything in in one dimension but then this is a general thing so your x could be in Rd in general which means that instead of line you are going to think of this as like a hyper plane that supports the function at this particular point that is what it means. So, now essentially in words what is the saying well I can the functions value is always greater than the linear approximation that you can get at any point.

So, this is nothing specific with x, I can take some other point now if I ask the same question if this was my x dash then the function the linear approximation of the function at x dash is going to look something like this. So, now this would be well for as a function of y this is going to be f(x) dash + y - x transpose del f(x) dash - x dash transpose del f(x) dash and so on.

So this now again you can see that the red line lower bounds the function itself, the blue line also lower bounds the function and no matter which point you take if you take the function the linear approximation of the function at that point it is going to lower bond the function and if this happens for every point then the function is convex so this is another alternative definition and which will be very useful maybe in proving certain things about certain machine learning problems also, okay.

So, what does this essentially, why what does this essentially mean? So, if you remember from our gradient descent type of a discussion, we argued that you know if you have a function which is f(x) + epsilon d we said that our Taylor series would say that this is something like f(x) + epsilon times d transpose del f(x) + some higher order terms.

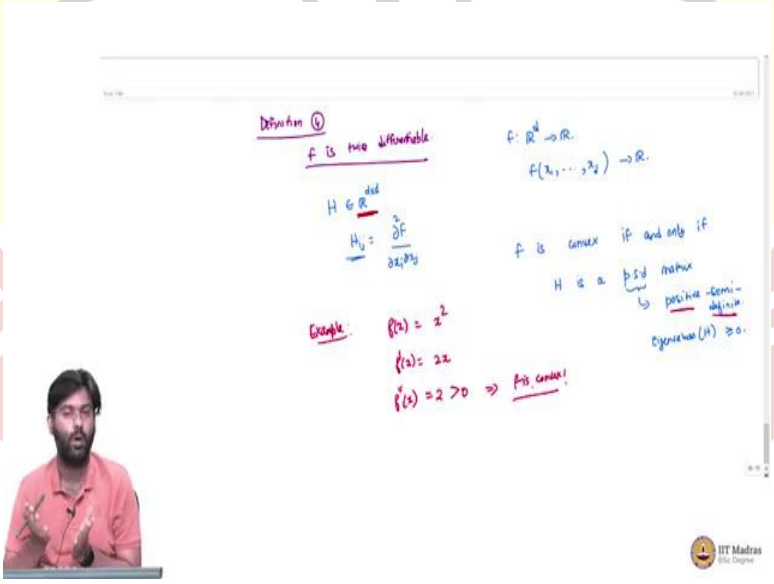
So and now, now I am going to think of let us say if I ignore the higher order terms so higher order terms and I am trying to approximate this function using only the linear terms then this approximate function is going to look at look like f(x) + epsilon d transpose del f(x), so it is just this part I have removed the other part now what is this telling us so this is what we this was the notation we used in gradient descent algorithms description.

Now imagine y is just x + epsilon d, so in this case just substitute y as x + epsilon d which means that if you want f of y, so f of y would be f(x) + well what would be epsilon d be; epsilon d because y is x + epsilon d epsilon d would simply be y - x. Now transpose del f(x) + higher order terms and now we are saying we are ignoring the higher order terms and then only thinking of this as a linear function at that point x.

Now, this is no longer going to be equal to because you are removed the higher order terms in gradient descents argument we said that this is like approximately equal to this for small enough epsilon and so on but then here something much more stronger holds. So, what we are saying in terms of convexity is that this is always greater than or equal to and we do not need epsilon to be small anything of that sort.

So, the functions value will be higher than the linear approximation no matter how far you go away from that point, so this blue line is going to lower bound this function in the entire space of Rd, in this case it is R so and if this happens for every point then we call that function also as a convex function. So, this is one other alternative way to think about convex functions.

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The final way that we will look at of thinking about convex functions and this is again definition-4 and in this particular definition we are going to assume f is not just differentiable but then twice differentiable. We are going to think of f as twice differentiable and we may or may not encounter this a lot but it is nevertheless good to understand this, so think of again f as a function from Rd to

R. Now I am going to define this matrix H which is a matrix in d cross d where H i j is simply the partial of think of this f as x 1 to x d to R, so that is what this function maps.

So, this is going to be the partial derivative of x i and x j, so this is going to be a matrix of double derivatives with respect to the corresponding i j entry and one can argue that f is convex if and only if something happens about this function H, about this matrix H. In particular the property of H is H is what is called as a PSD matrix which is stands for positive semi-definite it simply means that all Eigen values of H are greater than or equal to 0.

So, think of this as in some sense a curvature capturing the curvature of the function itself and intuitively this simply means that the function always should have a positive curvature, the moment it has a negative curvature it kind of goes above and comes back below and that is no longer a convex function. So, only if it has a positive curvature or a non-negative curvature, we will call it as a convex function.

So where is this useful? A simple example where this may be useful is as follows, so if you think of, f(x) as x square is the simple R to R function we already I think I mean it is not too hard to show that this is a convex function I mean there are several ways you can show this if you had to use this definition then you would do f dash of x now this would be that would be 2x.

Now, your H is a matrix in not Rd it is in a matrix in Rd cross d but then your d itself is 1 so it is just a one dimensional function so and that would be because and that value of H would simply be the f double dash of x and your f double dash of x in this case is 2 and the equivalent notion of positive semi definiteness but single numbers is just positive, it just has to be positive.

So, which means that this is greater than 0 which implies that f is convex. This is another way to show that a function x square is convex, I mean it is not the, it is one way to show x square is convex. You can show x square is convex using all four definitions that you have seen but this might be an interesting definition too okay, good.

So, we have four different definitions of convexity convex functions and all these would be useful in different situations. So, depending on what function you want to prove as convex or not you can use one of these definitions it might be easier to use one of these definitions than the others.